PROGRAM-6

1. Input Details

For each passenger:

- **Passenger Name**
- **Age**
- **Distance (km)**
- **Travel Class (Sleeper, AC 3-Tier, AC 2-Tier, First Class)**
- **Reservation Status**
- **Cancellation Status**
- ## 2. Distance-Based Fare Calculation
- Example fare slabs:

Distance (km)	Base Fare per km
Up to 100	₹1.00
101 - 500	₹0.90
Above 500	₹0.80

- **Base Fare**
- Base Fare = Distance × Rate per km

- ## 3. Travel Class Multipliers

Class	Multiplier
Sleeper	1.0
AC 3-Tier	1.5
AC 2-Tier	2.0
First Class	2.5

- **Class Fare**
- Class Fare = Base Fare × Class Multiplier

- ## 4. Age-Based Concessions
- Example policy:

Age Group	Concession
Below 5 Years	100%
5 - 12 Years	50%
60 Years & Above	40%
Others	0%

- **Discount**

- Discount $=$ Class Fare $\times \frac{\text{Concession}}{100}$

- **Fare After Concession**
- Net Fare $=$ Class Fare $-$ Discount

- ## 5. Reservation Fee

Class	Reservation Fee
Sleeper	₹20
AC 3-Tier	₹40
AC 2-Tier	₹50
First Class	₹60

- Ticket Amount $=$ Net Fare $+$ Reservation Fee

- ## 6. Cancellation Penalty
- Example rules:

Cancellation Time	Penalty
More than 48 hours before journey	10%
24 - 48 hours	20%
Less than 24 hours	50%

- **Refund Amount**
- Refund $=$ Ticket Amount $-$ Penalty

7. Algorithm

1. Input passenger details.
2. Determine fare rate using distance slab.
3. Calculate base fare.
4. Apply travel class multiplier.
5. Apply age-based concession.
6. Add reservation fee.
7. Calculate final ticket amount.
8. If cancelled, compute penalty and refund.
9. Generate fare summary for all passengers

Example

Passenger 1

- Age = 65
- Distance = 400 km
- Class = AC 3-Tier

Calculation

- Base Fare = 400 × ₹0.90 = ₹360
- Class Fare = 360 × 1.5 = ₹540
- Senior Citizen Discount (40%) = ₹216
- Net Fare = ₹324
- Reservation Fee = ₹40
- Ticket Amount = ₹364

Passenger 2

- Age = 30
- Distance = 400 km
- Class = Sleeper

Calculation

- Base Fare = ₹360
- Class Fare = ₹360
- Discount = ₹0
- Reservation Fee = ₹20
- Ticket Amount = ₹380

Output

Plain Text

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RAILWAY FARE STATEMENT
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Passenger 1 : ₹364
Passenger 2 : ₹380
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Total Fare : ₹744
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Programm

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# Railway Fare and Reservation Calculator

# Input number of passengers
n = int(input("Enter number of passengers: "))

distance = float(input("Enter travel distance (km): "))

# Distance-based fare
if distance <= 100:
    base_fare = distance * 1.00
elif distance <= 500:
    base_fare = 100 + (distance - 100) * 0.80
else:
    base_fare = 420 + (distance - 500) * 0.60

total_fare = 0

for i in range(1, n + 1):
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    print("\nPassenger", i)

    name = input("Enter passenger name: ")
    age = int(input("Enter passenger age: "))

    print("Travel Classes: 1. Sleeper  2. AC 3-Tier  3. AC 2-Tier")
    choice = int(input("Choose travel class (1-3): "))

    if choice == 1:
        travel_class = "Sleeper"
        class_multiplier = 1.0
        reservation_fee = 20
    elif choice == 2:
        travel_class = "AC 3-Tier"
        class_multiplier = 2.0
        reservation_fee = 40
    elif choice == 3:
        travel_class = "AC 2-Tier"
        class_multiplier = 3.0
        reservation_fee = 50
    else:
        print("Invalid class. Passenger skipped.")
        continue

    # Age-based concession
    if age < 5:
        concession_rate = 1.0
    elif age >= 60:
        concession_rate = 0.20
    else:
        concession_rate = 0.0

    passenger_base = base_fare * class_multiplier
    concession = passenger_base * concession_rate
    final_fare = passenger_base - concession + reservation_fee

    print("\n----- PASSENGER TICKET -----")
    print("Name:", name)
    print("Age:", age)
    print("Travel Class:", travel_class)
    print("Base Fare: Rs.", round(passenger_base, 2))
    print("Concession: Rs.", round(concession, 2))
    print("Reservation Fee: Rs.", reservation_fee)
    print("Final Fare: Rs.", round(final_fare, 2))

    total_fare += final_fare

print("\nTotal Reservation Fare: Rs.", round(total_fare, 2))

# Cancellation calculation
cancel = input("Do you want to calculate cancellation charges? (yes/no): ").lower()

if cancel == "yes":
    penalty_rate = float(input("Enter cancellation penalty percentage: "))
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    if 0 <= penalty_rate <= 100:
        penalty = total_fare * penalty_rate / 100
        refund = total_fare - penalty

        print("\n----- CANCELLATION DETAILS -----")
        print("Original Fare: Rs.", round(total_fare, 2))
        print("Cancellation Penalty: Rs.", round(penalty, 2))
        print("Refund Amount: Rs.", round(refund, 2))
    else:
        print("Invalid penalty percentage.")
```

output:

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Enter number of passengers: 54

Enter travel distance (km): 1100


Passenger 1

Enter passenger name: hanifa

Enter passenger age: 18

Travel Classes: 1. Sleeper  2. AC 3-Tier  3. AC 2-Tier

Choose travel class (1-3): 1


----- PASSENGER TICKET -----

Name: hanifa

Age: 18

Travel Class: Sleeper

Base Fare: Rs. 780.0

Concession: Rs. 0.0

Reservation Fee: Rs. 20

Final Fare: Rs. 800.0


Passenger 2

Enter passenger name: hafreen

Enter passenger age: 21

Travel Classes: 1. Sleeper  2. AC 3-Tier  3. AC 2-Tier

Choose travel class (1-3): 2
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----- PASSENGER TICKET -----

Name: hafreen

Age: 21

Travel Class: AC 3-Tier

Base Fare: Rs. 1560.0

Concession: Rs. 0.0

Reservation Fee: Rs. 40

Final Fare: Rs. 1600.0

Passenger 3

Enter passenger name: swathi

Enter passenger age: 17

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 3

----- PASSENGER TICKET -----

Name: swathi

Age: 17

Travel Class: AC 2-Tier

Base Fare: Rs. 2340.0

Concession: Rs. 0.0

Reservation Fee: Rs. 50

Final Fare: Rs. 2390.0

Passenger 4

Enter passenger name: lekhana

Enter passenger age: 16

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 5

Invalid class. Passenger skipped.

Passenger 5

Enter passenger name: sneha

Enter passenger age: 20

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 4

Invalid class. Passenger skipped.

Passenger 6

Enter passenger name: mastanbi

Enter passenger age: 54

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 4

Invalid class. Passenger skipped.

Passenger 7

Enter passenger name: babji

Enter passenger age: 65

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 5

Invalid class. Passenger skipped.

Passenger 8

Enter passenger name: rihana

Enter passenger age: 12

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 6

Invalid class. Passenger skipped.

Passenger 9

Enter passenger name: basith ali

Enter passenger age: 19

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 7

Invalid class. Passenger skipped.

Passenger 10

Enter passenger name: harika

Enter passenger age: 87

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 7

Invalid class. Passenger skipped.

Passenger 11

Enter passenger name: mounika

Enter passenger age: 45

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 8

Invalid class. Passenger skipped.

Passenger 12

Enter passenger name: honey

Enter passenger age: 5

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 9

Invalid class. Passenger skipped.

Passenger 13

Enter passenger name: mani

Enter passenger age: 78

Travel Classes: 1. Sleeper 2. AC 3-Tier 3. AC 2-Tier

Choose travel class (1-3): 8